

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and optimized ML compilers using **PyTorch** and **C++** for a novel NPU architecture, achieving a **22% reduction** in model compilation time.
- Designed and implemented advanced **quantization techniques** for deep learning models, improving inference efficiency by **30%** while maintaining **99% accuracy**.
- Optimized machine learning workloads for deployment on diverse hardware accelerators, resulting in a **15% increase** in average inference speed and a **25% decrease** in power consumption.
- Collaborated with strategic customers to integrate and tailor ML solutions, successfully resolving **complex performance challenges** and delivering impactful results for energy-efficient ML technology.
- Researched and prototyped innovative solutions in **ML compiler design** and system-level optimizations, contributing to the development of the world's lowest-power ML solution.
- Evaluated and debugged ML performance bottlenecks in complex workflows, creating thorough **technical documentation** that accelerated team knowledge sharing and problem-solving by **40%**.

SKILLS

ML Engineering – PyTorch, C++, ML Compilers, Quantization, ML Workloads, Performance Optimization, System-Level Optimizations, Hardware Accelerators, NPU Architecture

Programming & Scripting – Python, C++, Java, SQL, Shell Scripting

Data Analysis & BI – Power BI, DAX, Tableau, R, pandas, NumPy, scikit-learn, Data Modelling, Data Warehousing

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, Linux

Enterprise Systems – ERP Support, REST APIs, Change Management, Production Support, Incident Response, Splunk, Dynatrace

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India

01/2021

PROJECTS

NPU Compiler Optimization Framework (Qualcomm)

- Engineered a comprehensive framework for optimizing ML compilers on Qualcomm's NPU architecture using **PyTorch** and **C++**, reducing model compilation cycles by **28%**.
- Implemented novel **quantization algorithms** within the framework, achieving a **35% improvement** in model efficiency and a **10% reduction** in memory footprint for client-deployed models.

- Developed automated performance evaluation pipelines that identified and resolved inference bottlenecks, increasing overall workload throughput by **20%** and documenting findings for customer integration.

Automated Payment Transaction Monitoring System (HCL Technologies)

- Developed Python scripts to automate the monitoring of payment transaction flows within a financial-services dashboard, identifying and alerting on failed transactions with **99.5% accuracy**.
- Integrated with **Splunk** for real-time log analysis and Dynatrace for application performance monitoring, reducing P1 incident resolution time by **25%**.
- Designed and implemented SQL queries across PostgreSQL and Oracle databases for transaction data validation and reconciliation workflow support, ensuring data integrity for critical financial operations.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)